

Systematic Risk Assessment and Quantification for Autonomous Robotic Construction on Mars

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ABSTRACT

This paper investigates the risks associated with autonomous robotic construction on Mars, emphasizing a comprehensive approach to identifying, assessing, and mitigating challenges unique to extreme extraterrestrial environments. Based on a systematic review of literature from 2014 to 2024, the study focuses on five critical risk areas: atmospheric conditions, topography, energy and power requirements, communication systems, and equipment reliability. These categories inform a quantitative risk assessment method that maps risks by likelihood and impact, supporting the prioritization of mitigation strategies. Key hazards, such as dust contamination, thermal inertia, and reduced mobility, are shown to significantly affect equipment performance, energy supply, and communication systems. The study offers practical insights for developing resilient, efficient construction processes suited to Martian conditions, with broader relevance to construction in remote and harsh terrestrial environments. Future work will incorporate expert feedback to refine the proposed frameworks and mitigation strategies.

INTRODUCTION

NASA's goal of establishing a sustained human presence on Mars requires the development of innovative construction methods capable of addressing the planet's unique and extreme conditions. The Martian environment presents challenges far beyond those of terrestrial construction, including radiation exposure, drastic temperature fluctuations, micrometeorite impacts, and seismic risks. These are further complicated by resource limitations, unfamiliar terrain, and the operational complexities of autonomous systems, such as communication delays and multi-robot coordination (Gheorghiu et al. 2016; Shojaei et al. 2018). These factors necessitate novel construction strategies adapted specifically for extraterrestrial applications.

Advancements in Martian habitat construction have addressed various technological and methodological areas, including autonomous robotics, in-situ resource utilization, and additive manufacturing. For instance, NASA's Centennial Challenges program has fostered innovation through competitions focused on regolith excavation, space robotics, and 3D-printed habitats (Yashara et al. 2019; Halbach 2017). Proposed habitat designs include terrestrial-style structures using local materials, inflatable bases, and mobile modular units (Knez & Khalilidermani 2021).

Despite progress, risks tied to implementing these systems in extreme environments remain underexplored, especially from a construction management perspective (Nuaimi et al. 2020).

A prior systematic review conducted by the authors identified major risks specific to autonomous construction on Mars. While the review offered a broad understanding of potential hazards, it primarily focused on risk identification without detailing quantification or mitigation strategies. This gap highlights the need for actionable frameworks that link theoretical analysis with practical construction planning. Building on these findings, this study evaluates and prioritizes risks through a structured framework using a 5x5 risk matrix. This matrix assesses hazards based on likelihood and impact, forming the basis for mitigation strategies. The results offer practical guidance for planning autonomous construction in extreme environments, with implications for Martian missions and similar challenges in remote or harsh terrestrial regions (Naser & Chehab 2018).

LITERATURE REVIEW

The development of autonomous robotic construction systems for Mars presents unprecedented challenges due to the planet's harsh environment and complex terrain. This literature review examines the current state of knowledge on critical risk areas and identifies gaps in addressing these risks comprehensively. Key focus areas include atmospheric conditions, geological and topographical hazards, equipment limitations, and energy demands.

Atmospheric Conditions and Environmental Challenges

Mars presents an environment marked by extreme atmospheric conditions, including significant temperature fluctuations, low atmospheric pressure, and pervasive dust storms. The average surface temperature is -81°F , with extremes from -195°F to 70°F , posing severe operational challenges for robotic systems. Frequent dust storms—lasting weeks or even months—reduce visibility, impair equipment, and degrade solar power efficiency (Wan et al. 2024). The thin, carbon dioxide-rich atmosphere provides little protection against cosmic radiation and solar winds, which can disrupt electronics and communication (Liu et al. 2022). Seasonal variations further influence environmental risks; temperature shifts caused by thermal inertia and dust activity affect atmospheric pressure gradients, worsening dust abrasion and equipment wear. These conditions underscore the need for robust robotic systems capable of enduring highly variable and adverse climates.

Geological and Topographical Hazards

Mars's rugged and varied terrain, including craters, boulder fields, steep slopes, and loose regolith, poses significant obstacles to the mobility and stability of autonomous systems. Rovers from past missions, such as Curiosity, experienced substantial wheel damage due to rough terrain and limited traction control (Stimpson et al. 2017). Loose regolith and uneven surfaces increase the risk of wheel slippage and sinkage, disrupting construction operations and delaying timelines. Subterranean environments like lava tubes and caves have been proposed as potential habitats for their natural shielding from radiation and temperature extremes, but navigating them is challenging due to poor visibility, unstable ground, and restricted access (Olsen & Alexis 2023). Additional hazards, including landslides, marsquakes, and high-speed meteorite impacts, further complicate construction planning and risk mitigation on the Martian surface.

Equipment Limitations and Durability

The durability of robotic systems in extraterrestrial environments remains a key concern. Dust contamination poses mechanical and electronic risks, as fine particles can infiltrate joints, seals, and sensitive components, reducing both efficiency and lifespan (Zanon et al. 2023). Accumulated dust on solar panels also decreases energy generation, intensifying power management challenges during long missions (Wan et al. 2024). Communication delays between Mars and Earth require highly autonomous systems capable of performing tasks with minimal human input. To support this, advanced algorithms and decision-making frameworks are essential for real-time responses to dynamic conditions and unforeseen obstacles (Bonafini & Sacchi 2020).

Energy Requirements and Power Limitations

Power generation and storage are central challenges for sustaining long-term construction activities on Mars. Solar power, the most commonly used energy source for robotic missions, is less effective on Mars due to reduced solar irradiance and frequent dust storms that obscure sunlight. Solar arrays also face efficiency losses due to temperature extremes and dust accumulation (Wan et al. 2024). While nuclear power offers a more reliable alternative, it introduces additional logistical and safety concerns (Alkhawashke et al. 2022). Proposed solutions include hybrid energy systems that combine solar and nuclear power, as well as innovations in energy storage technologies such as thermoelectric generators and advanced lithium-ion batteries (Salinas-Camus et al. 2023). However, these technologies require further development and testing to ensure their reliability under Martian conditions.

Knowledge Gaps and Future Opportunities

While previous research has identified numerous risks related to Martian construction, it has often addressed these challenges in isolation rather than through an integrated framework. This study seeks to bridge that gap by applying a quantitative risk assessment approach to evaluate and prioritize construction hazards. By combining insights from existing literature with expert evaluations, it presents a structured framework to identify, categorize, and address key challenges. The findings offer actionable strategies for planning and executing autonomous construction in extreme extraterrestrial environments.

METHODOLOGY

This study employs a structured methodology to identify, categorize, assess, and mitigate risks associated with autonomous robotic construction on Mars and other extreme extraterrestrial environments. Building upon a previously conducted systematic literature review (SLR), the study expands its foundational findings by developing actionable risk assessment and mitigation frameworks. The SLR, currently under review, analyzed over 2,000 publications and categorized risks into five key areas: Atmospheric Conditions, Topography, Energy and Power Requirements, Communication Systems, and Equipment Reliability. After a rigorous screening and full-text review of 100 selected articles, qualitative analysis was conducted to systematically organize risks. These were then grouped within the 3M framework, Mother Nature

(environmental), Machine (technological), and Material (resource-related) (Schuldt et al. 2021)—to clarify their sources and interdependencies (see Figure 1).

To advance beyond the SLR, this study introduces a quantitative approach to prioritize risks. A 5x5 risk matrix was used to assess the likelihood and impact of each identified risk, classifying them into high, medium, and low priority. High-priority risks received focused attention, as they pose the most significant threats to construction feasibility. The quantitative analysis also evaluated hazards across major Martian construction activities, including Site Exploration and Selection, Site Preparation, Substructure Construction, Superstructure Construction, and Finishing. For each activity, hazard likelihood and impact were assessed, and risk factors calculated, offering a focused framework for understanding risk across construction stages.

RESULTS, ANALYSIS, AND DISCUSSION

Risk Assessment

Risk assessment is a critical component of project management, particularly in highly uncertain environments such as Martian construction. It involves evaluating the probability and impact of potential risks to develop effective strategies for mitigation and management (Grgurevic 2024). Various approaches to risk assessment exist, each with its own strengths and limitations (Björnsdóttir et al. 2022). While many of these methods, including qualitative assessments, are inherently subjective, they remain valuable tools for ranking and prioritizing risks in complex scenarios, thereby forming a foundation for effective risk management strategies (Elmontsri, 2014; Florig et al. 2001; Kindinger & Darby 2000).

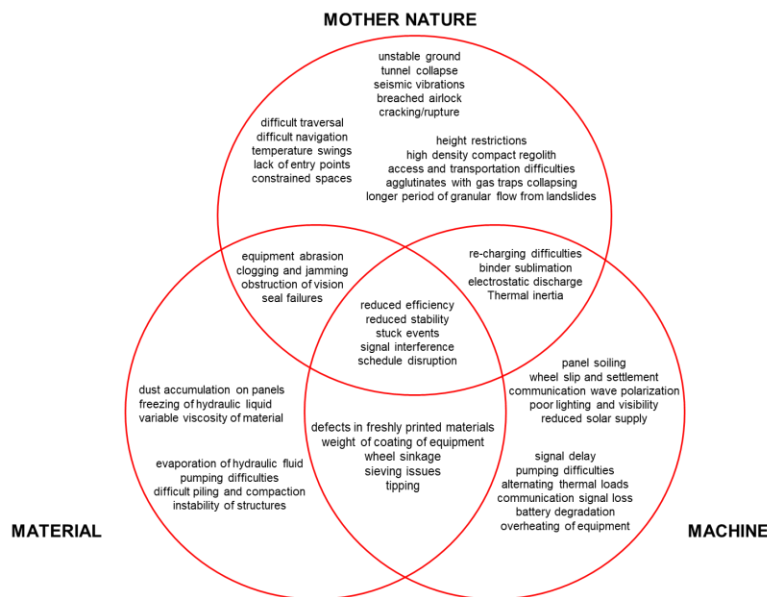


Figure 1. Classification of risks based on the 3M framework

Among the tools available, the 5x5 risk matrix, originally derived from the American Military Standard (MIL-STD-882B), is widely recognized for its applicability in diverse contexts (Florig et al. 2001). This matrix facilitates the detailed evaluation of risks by

categorizing them across a broad range of potential consequences and likelihoods, making it particularly useful in environments characterized by uncertainty and complexity, such as Martian construction. In this study, we employed the 5x5 risk matrix to systematically evaluate the risks associated with autonomous robotic construction on Mars. The matrix's robust framework enables the categorization of risks based on both the likelihood of their occurrence and the severity of their potential impact, providing a structured approach to prioritize and manage these risks.

The authors rated each identified risk using information gained through an extensive review of the literature and relevant papers. These ratings, reflecting both the frequency and severity of each risk according to Tables 1 and 2, respectively, were then mapped onto the 5x5 risk matrix, as shown in Figure 2, to determine the overall risk level. This approach enabled the quantification and prioritization of risks, thereby guiding the development of targeted mitigation strategies critical to the success of Martian construction.

The detailed results of the risk assessment categorize hazards into key themes: atmospheric conditions, topographical challenges, power supply constraints, and equipment functionality. The matrix identifies several high-risk hazards requiring immediate attention and mitigation strategies. For instance, dust accumulation on solar panels and equipment, scoring 25 in the 5x5 matrix, underscores the critical threat posed by Martian dust to energy efficiency and equipment functionality. This highlights the necessity for advanced cleaning technologies or protective coatings. Similarly, reduced traction and difficult terrain traversal, particularly in loose regolith and craters, received high-risk ratings, emphasizing the need for robust mobility solutions such as adaptive suspension systems or innovative wheel designs.

Table 1. Risk Likelihood Rating Criteria

Rating	Likelihood	Description
5	Definite	Is expected to frequently occur
4	Highly Probable	Has a good chance of occurring and is not unusual
3	Possible	Likely to occur sometime
2	Negligible	Has a slight chance of occurring
1	Remote	Not impossible but only in exceptional circumstances

Table 2. Risk Severity Rating Criteria

Rating	Severity	Description
5	Catastrophic/ Disastrous	Irreversible failure leading to mission termination or complete system loss
4	Major	Severe disruption with prolonged downtime affecting critical mission objectives
3	Moderate	Noticeable impairment causing partial loss of capabilities
2	Minor	Limited impact with manageable reduction in performance
1	Insignificant	Minimal effect with quick recovery and no lasting impact

Moderate risks, such as communication signal interference and thermal inertia, represent ongoing challenges that, while less severe, demand tailored strategies like redundant communication systems and thermally resilient materials. These measures ensure operational

stability without diverting substantial resources from high-priority hazards. Low-risk items, including clogging of equipment and seal failures, reflect nuanced operational challenges. Although these risks may not pose an immediate threat, they could escalate under specific circumstances. Continuous monitoring and adaptive designs are essential to mitigate these risks effectively.

Likelihood	1	5 (Medium)	10 (Medium)	15 (High)	20 (High)	25 (High)
	2	4 (Low)	8 (Medium)	12 (Medium)	16 (High)	20 (High)
	3	3 (Low)	6 (Medium)	9 (Medium)	12 (Medium)	15 (High)
	4	2 (Low)	4 (Low)	6 (Medium)	8 (Medium)	10 (Medium)
	5	1 (Low)	2 (Low)	3 (Low)	4 (Low)	5 (Medium)
		1	2	3	4	5
Severity						

Figure 2. 5 X 5 Risk Matrix

Overall, the matrix demonstrates that the most significant threats arise from the interaction of environmental factors, such as dust storms and extreme temperatures, and their impact on critical systems like energy supply, mobility, and structural integrity. By categorizing risks across varying levels of urgency, the matrix provides a structured roadmap for targeted mitigation strategies, ensuring high-risk factors are addressed promptly while maintaining awareness of evolving moderate and low-level risks. These findings not only inform the design and planning of autonomous construction projects on Mars but also offer valuable insights for analogous challenges on Earth, such as operations in remote or extreme environments. This analysis underscores the importance of developing resilient, efficient construction processes capable of overcoming the unique challenges posed by extraterrestrial environments.

Risk Factor Analysis

The Risk Factor Analysis (RFA) method, developed by Kindinger and Darby (2000), provides a systematic and qualitative approach for evaluating and prioritizing project risks across key dimensions such as schedule, cost, and technical performance. It enables the ranking of risks by specific activities and forms a foundation for quantitative modeling by synthesizing insights from the risk assessment process. The table in Appendix 2 aligns identified risk factors with construction activities—Site Exploration and Selection, Site Preparation, Substructure Construction, Superstructure Construction, and Finishing.

The scoring system quantifies each risk based on its prevalence and impact within these activities. It provides cumulative scores for each risk factor and for each construction phase. Each risk score reflects potential challenges posed by specific hazards during the associated activities, while each activity's cumulative score indicates the total impact of all risks during that phase. This supports clear prioritization of mitigation efforts based on the relative severity and complexity of the challenges. This scoring framework enables systematic risk assessment across all phases, guiding resource allocation toward high-priority activities and supporting tailored mitigation strategies.

Figure 3 hierarchically ranks risks by RFA scores, highlighting the most critical to the least concerning within the full construction process. Figure 4 ranks construction activities by cumulative risk factors from all identified hazards. Together, these figures provide a comprehensive perspective for prioritizing risk management efforts by targeting both significant risks and the most vulnerable construction phases. Figure 3 identifies "Equipment damage" and "Equipment abrasion; wear and tear" as the most significant risks, reflecting the harsh conditions equipment will face on Mars. These findings underscore the need for robust, resilient equipment design and maintenance strategies to ensure reliability and longevity. "Unstable ground" and "Difficult traversal" are also major concerns, consistent with the challenges of Mars's uneven terrain. These risks emphasize the importance of thorough site assessment, preparation, and advanced mobility solutions.

The prominence of "Dust accumulation on equipment" and "Dust storms" supports earlier analysis identifying dust as a pervasive hazard. Dust negatively affects equipment performance, visibility, and communication, highlighting the need for mitigation technologies such as protective coatings, filtration systems, and routine maintenance.

Lower-ranked risks, while less immediate, still present challenges—especially when moderate risks accumulate. For instance, "Gantry system installation and dismantling difficulties" and "Variable viscosity of material" reflect technical complexities in 3D printing and structural assembly, requiring detailed planning, testing, and adaptable design.

According to Figure 4, "Substructure construction" carries the highest total risk, due to challenges in foundation stability, dust contamination, and temperature extremes. "Superstructure construction" and "Site preparation" also show substantial risk levels, reflecting the technical and logistical complexities of 3D printing and equipment mobilization on rough terrain. "Site Exploration and Selection" shows lower total risk, mainly due to accessibility and terrain evaluation in potential subterranean sites. "Finishing" ranks lowest, consistent with its less hazardous and complex nature.

The insights from this analysis are critical for guiding risk management in Martian construction. By focusing on top-ranked risks, especially those involving equipment durability and terrain, planners can allocate resources more effectively and develop strategic mitigation plans. These findings offer essential guidance for ensuring the success of construction activities in Mars's challenging environment.

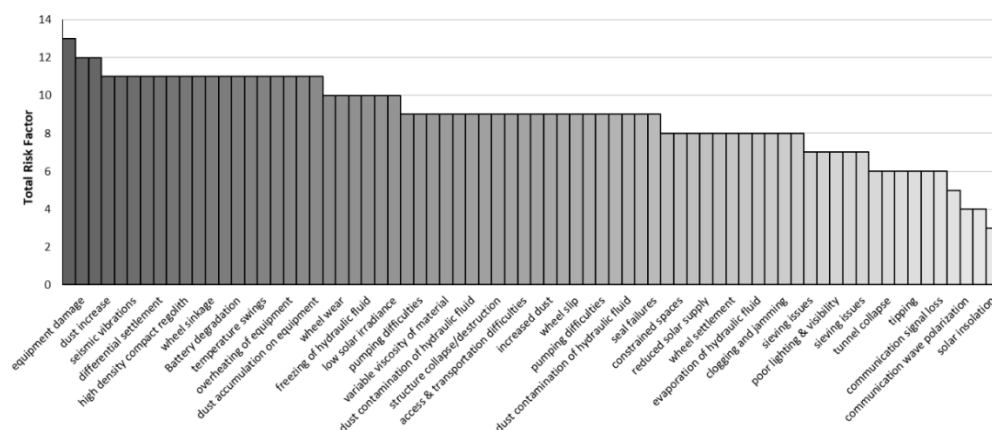


Figure 3. Total Risk Factor for Identified Risks

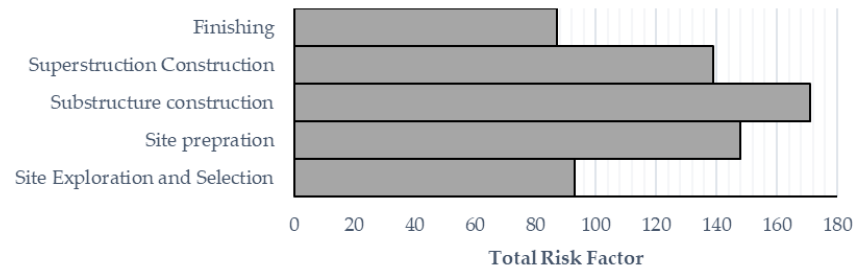


Figure 4. Total Risk Factor for Martian Construction Phases

CONCLUSION

This study systematically assessed the risks associated with autonomous robotic construction on Mars by employing a risk matrix and Risk Factor Analysis (RFA). By categorizing and quantifying risks across various construction phases, the study provides a structured framework to prioritize hazards and guide the development of targeted mitigation strategies. The findings contribute to the growing body of knowledge in extraterrestrial construction, offering insights that can be applied to analogous challenges on Earth. Future research should focus on validating the identified risks and their prioritization through expert reviews and collaborative workshops. Additionally, practical simulations and prototype testing in controlled environments that mimic Martian conditions will be essential for refining the proposed risk management strategies and ensuring their effectiveness in real-world applications. This iterative validation process will enhance the robustness and applicability of the findings,

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